

Worksheet: 2-Input, 3-Hidden Node Neural Network

SECTION 1: Question Setup & Parameters

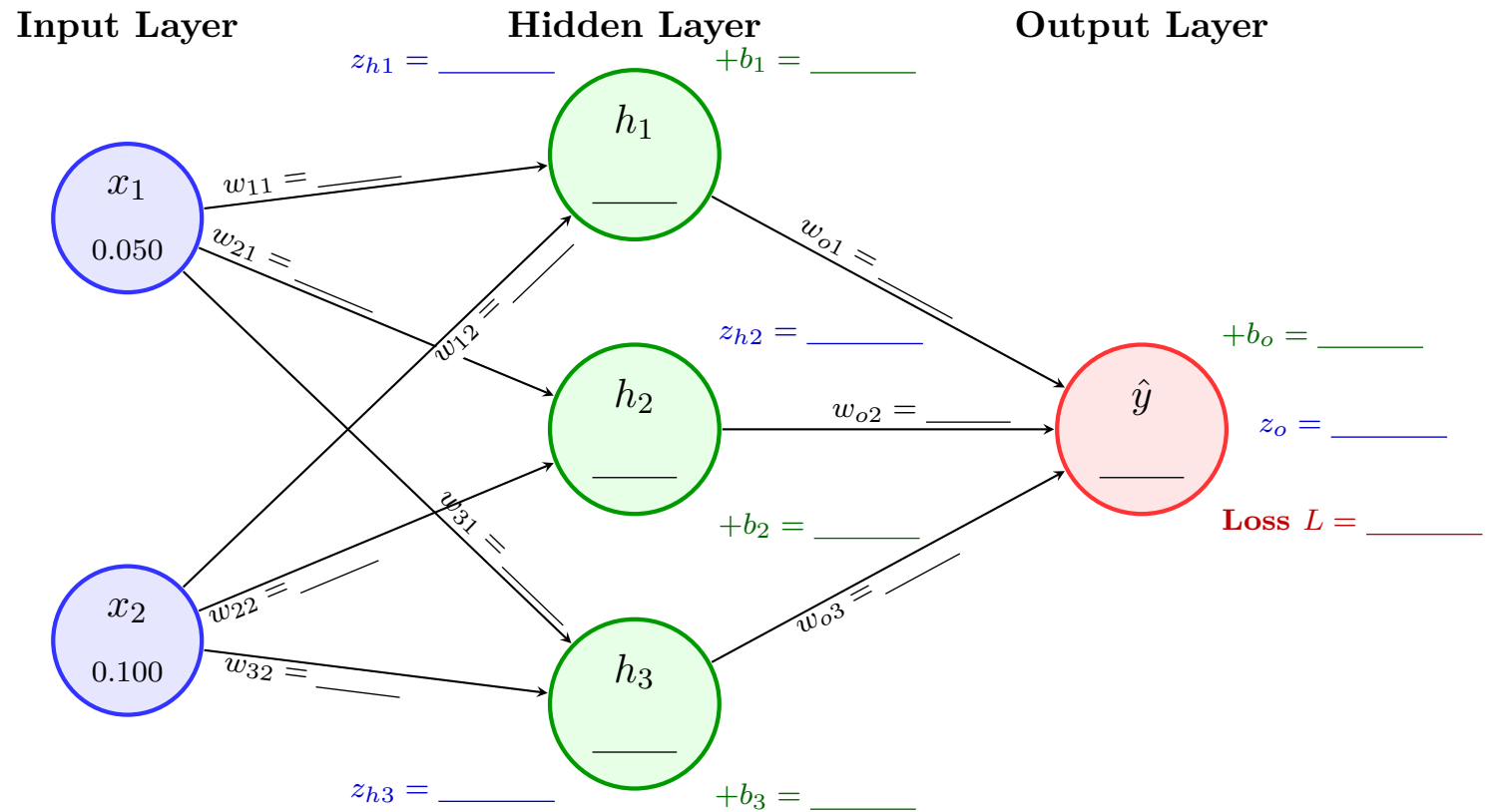
Problem Setup & Global Parameters:

- **Input Vector:** $\mathbf{x} = [x_1, x_2]^T = [0.050, 0.100]^T$
- **Target Output:** $y = 1.000$
- **Learning Rate:** $\eta = 0.500$
- **Activation Function:** Sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$
- **Activation Derivative:** $\sigma'(z) = \sigma(z)(1 - \sigma(z))$
- **Loss Function:** Squared Error $L = \frac{1}{2}(y - \hat{y})^2$
- **Loss Derivative:** $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$
- **Calculation Precision:** Fix all intermediate and final calculations at **4 decimal places** by truncation (cut off after 4th decimal digit without rounding).

Initial Network Weights & Biases:

- **Input $\rightarrow$ Hidden Weights ($W^{(1)}$):**
 $w_{11} = 0.1500, \quad w_{12} = 0.2000$
 $w_{21} = 0.2500, \quad w_{22} = 0.3000$
 $w_{31} = 0.3500, \quad w_{32} = 0.4000$
- **Hidden Biases:** $b_1 = 0.3500, \quad b_2 = 0.3500, \quad b_3 = 0.3500$
- **Hidden $\rightarrow$ Output Weights ($W^{(2)}$):**
 $w_{o1} = 0.4500, \quad w_{o2} = 0.5000, \quad w_{o3} = 0.5500$
- **Output Bias:** $b_o = 0.6000$

Neural Network Architecture Diagram (Initial Weights & First Forward Pass)



SECTION 2: First Forward Pass Calculations

2.1 Hidden Node Net Inputs (z_{hi}) and Activations (h_i):

- (a) $z_{h1} =$ _____ $\Rightarrow h_1 =$ _____
- (b) $z_{h2} =$ _____ $\Rightarrow h_2 =$ _____
- (c) $z_{h3} =$ _____ $\Rightarrow h_3 =$ _____

2.2 Output Unit Prediction ($\hat{y}$) and Initial Loss (L):

- (a) $z_o =$ _____ $\Rightarrow \hat{y} =$ _____
- (b) Initial Loss $L =$ _____

SECTION 3: Backward Pass & Update ALL Weights and Biases ($\eta = 0.500$)

3.1 Output Layer Error Gradient (δ_o):

- (a) $\delta_o =$ _____

3.2 Output Layer Updates (w_{oi}^+ and b_o^+):

- (a) $w_{o1}^+ =$ _____
- (b) $w_{o2}^+ =$ _____
- (c) $w_{o3}^+ =$ _____
- (d) $b_o^+ =$ _____

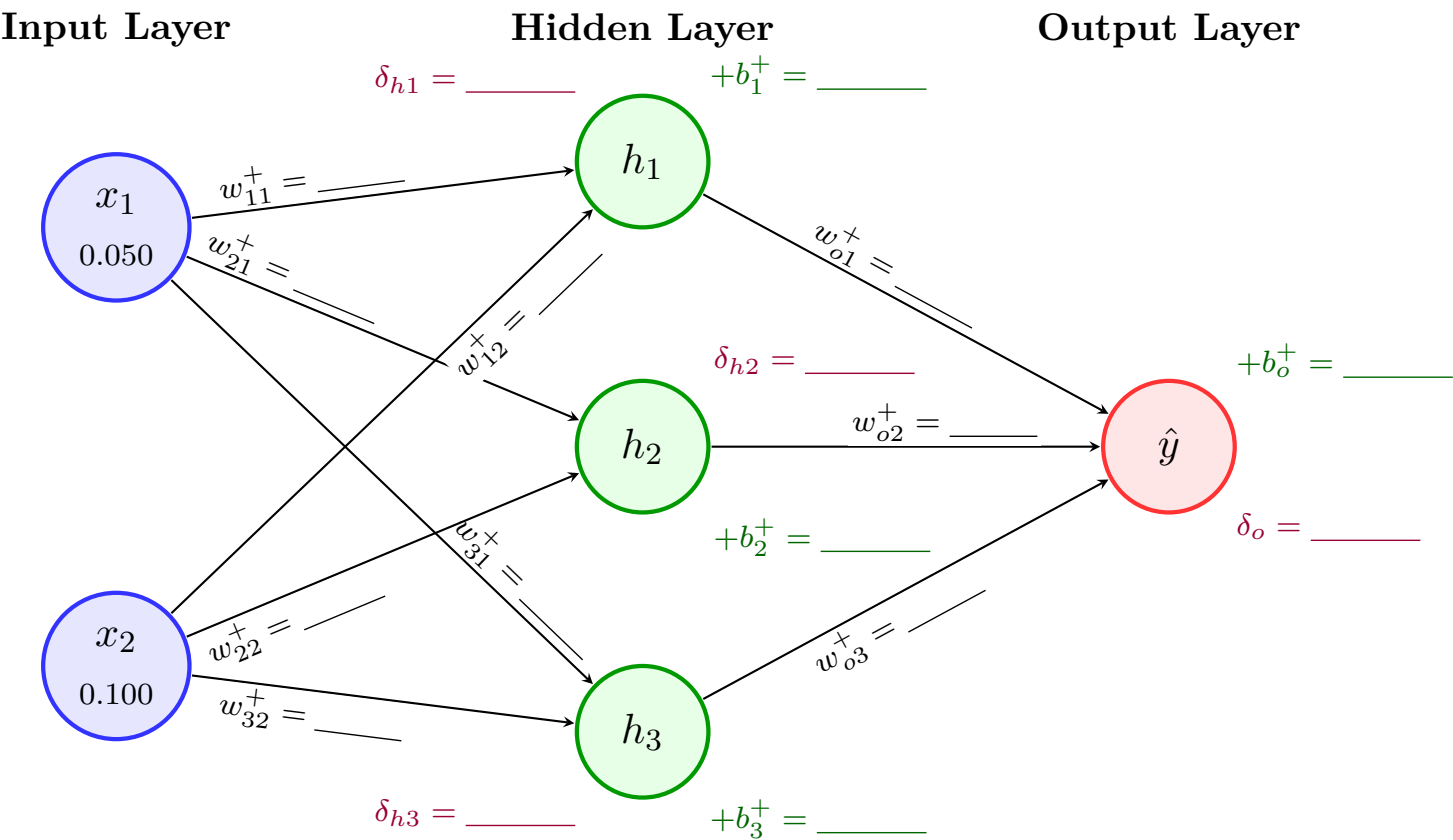
3.3 Hidden Layer Error Gradients (δ_{hi}):

- (a) $\delta_{h1} =$ _____
- (b) $\delta_{h2} =$ _____
- (c) $\delta_{h3} =$ _____

3.4 Input-to-Hidden Layer Updates (w_{ij}^+ and b_i^+):

- (a) $w_{11}^+ =$ _____
- (b) $w_{12}^+ =$ _____
- (c) $w_{21}^+ =$ _____
- (d) $w_{22}^+ =$ _____
- (e) $w_{31}^+ =$ _____
- (f) $w_{32}^+ =$ _____
- (g) $b_1^+ =$ _____
- (h) $b_2^+ =$ _____
- (i) $b_3^+ =$ _____

Updated Neural Network Architecture (Gradients & New Weights/Biases)



SECTION 4: Second Forward Pass (Verification of Learning)

4.1 Recalculate Hidden Activations (h_i^+):

- (a) $z_{h1}^+ = \underline{\hspace{2cm}} \Rightarrow h_1^+ = \underline{\hspace{2cm}}$
- (b) $z_{h2}^+ = \underline{\hspace{2cm}} \Rightarrow h_2^+ = \underline{\hspace{2cm}}$
- (c) $z_{h3}^+ = \underline{\hspace{2cm}} \Rightarrow h_3^+ = \underline{\hspace{2cm}}$

4.2 Recalculate Output Prediction ($\hat{y}^+$) and New Loss (L^+):

- (a) $z_o^+ = \underline{\hspace{2cm}} \Rightarrow \hat{y}^+ = \underline{\hspace{2cm}}$
- (b) New Loss $L^+ = \underline{\hspace{2cm}}$
- (c) Has the loss decreased after gradient descent update? ($L^+ < L$): $\underline{\hspace{2cm}}$

Neural Network Architecture Diagram (Second Forward Pass with Updated Weights)

